Aran Learn 28: Dashboards and Metrics

For: everyone. Inbathamizhan first (the dashboard UI), Aasif for the numbers behind it. **Time:** about 1 hour with practice. **Before this:** Learn 4 (probability for detection), Learn 17 (wiring a design to a live API) and Learn 26 (SaaS).

A bank buying Aran will open the Aran Cloud dashboard every morning. It must answer three questions in five seconds: **Is Aran working? Is it protecting us? Is it slowing payments?** This doc explains how a dashboard gets its numbers and how to show them honestly.

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1. Metrics, events and logs

Kind	What it is	Example
Event	One thing that happened, with details	"Txn TXN-B-123 protected 3 values in 2.1 ms"
Metric	A number added up over time	"4,812 transactions protected in the last hour"
Log	Free text written by a program	`INFO protect ok tenant=t_b7 ms=2.1`

Aran Cloud stores **events** (`usage_events` , one row per call) and works out **metrics** from them when the dashboard asks. At large scale you keep metrics pre-counted in time buckets instead (section 3).

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2. The four kinds of metric

- **Counter**: only goes up. Transactions processed, values tokenised. You usually show its **rate**: per minute, per day.
- **Gauge**: goes up and down. Open alerts, keys alive, queue length.
- **Ratio**: one count over another. **Coverage** = values tokenised ÷ values found. **Error rate** = failed calls ÷ all calls.
- **Distribution**: many values, like latency. You show **percentiles**, not the average.

Why percentiles, not averages

Ten payments take 2, 2, 2, 2, 2, 2, 2, 2, 2 and 80 ms. The **average** is 9.8 ms, which describes none of them. The **median (p50)** is 2 ms: a normal payment. The **p95** is the time 95% of payments beat; it shows the slow tail. Banks write SLAs on p95 or p99.

```
import numpy as np
lat = [2, 2, 2, 2, 2, 2, 2, 2, 2, 80]
np.percentile(lat, 50), np.percentile(lat, 95)   # (2.0, 44.9)
```

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3. Time buckets

To draw "transactions per minute for the last hour", count the events in each one-minute **bucket**:

```
SELECT CAST(at / 60 AS INTEGER) * 60 AS minute, COUNT(*) AS n
FROM usage_events
WHERE tenant_id = ? AND at > strftime('%s','now') - 3600
GROUP BY minute ORDER BY minute;
```

Two traps: - **Empty buckets.** A minute with no traffic returns no row. Fill it with 0, or the chart draws a line straight across the gap and lies. - **The current bucket is not finished.** The last minute always looks low. Mark it as "in progress" or leave it out.

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4. What the Aran Cloud dashboard should show

Question	Metric	Kind
Is it working?	Calls per minute, error rate, last call time	Counter rate, ratio
Is it protecting us?	Values tokenised, coverage %, raw values passed, by type (Aadhaar, PAN, UPI id, card)	Counters, ratio
Is it slowing payments?	Added latency p50 / p95 per minute against the budget	Distribution
Where does data go?	Values per destination (logs, fraud, analytics, AI)	Counters by label
Are we compliant?	Erasure receipts, detokenise reads by person, audit chain verified	Counters, gauge
What does it cost?	Transactions this month against the plan	Counter vs. limit

A **label** (or dimension) splits a metric: transactions by **channel** (UPI, card…), by **destination**, by **bank**. Keep labels to a small fixed list; never use a customer id as a label.

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5. Show numbers honestly

- **Every number says its window.** "4,812 (last hour)", not "4,812".
- **Every chart starts its y-axis at 0** for counts, so a small change does not look huge.
- **Say when there is no data.** "No transactions yet" is not the same as "0 protected".
- **Say how fresh it is.** "Updated 3 s ago"; when the server is unreachable, say so instead of freezing old numbers (Learn 25, section 4).
- **Colour means status, not decoration.** Green = within target, amber = close, red = broken. Never colour alone: add an icon or word for colour-blind readers.
- **One tenant only.** The dashboard of Bank B must never include Bank A's numbers (Learn 26).

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6. SLIs, SLOs and alerts

- **SLI** (service level indicator): a metric that measures what customers feel. Example: p95 added latency.
- **SLO** (objective): the target. "p95 added latency under 10 ms, 99.9% of minutes this month."
- **Error budget**: how much you may miss. 0.1% of a month is about 43 minutes.
- **Alert**: tells a person when the SLO is in danger. Alert on symptoms ("p95 over 10 ms for 5 minutes"), not on causes ("CPU 80%").

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7. Live updates

The dashboard can get fresh numbers in three ways:

Way	How	Good for
Polling	Ask every few seconds	Simple; what Aran does today
Server-Sent Events	The server keeps one connection open and pushes updates	Live counters and feeds
WebSocket	Two-way open connection	Chat, collaborative tools

Polling every 2 to 5 seconds is enough for a bank dashboard. Stop polling when the tab is hidden.

Exercises

1. Latencies (ms): 1, 1, 2, 2, 2, 3, 3, 4, 9, 60. Work out the average, p50 and p95. Which would you put in an SLA?
2. Run the bucket query above on `aran_cloud/cloud.db` for your workspace. Which minutes are missing, and how will you fill them?
3. Bank B processed 1,200 transactions; Aran found 3,100 identifiers and tokenised 3,094. What is the coverage, and what would you show next to it?
4. Your SLO is 99.9% of minutes under 10 ms. In a 30-day month, how many bad minutes can you have?

Words to know

Word	Meaning
Event	One thing that happened, stored with its details.
Metric	A number summed or measured over time.
Counter	A metric that only goes up.
Gauge	A metric that goes up and down.
Rate	A counter's change per unit of time.
Percentile (p50, p95)	The value that 50% or 95% of measurements are below.
Bucket	A fixed slice of time that events are counted in.

Word	Meaning
Label (dimension)	A field that splits a metric, like channel or destination.
Coverage	The share of found identifiers that were protected.
SLI / SLO	What you measure / the target you promise.
Error budget	How much the SLO may be missed.
Server-Sent Events	A one-way open connection the server pushes updates over.